



पंचायती राज मंत्रालय
Ministry of Panchayati Raj



IIT Tirupati
Navavishkār
I-Hub Foundation



National AI/ML Hackathon

Final Round Presentation

Team ID: DTM-AI-2026

Organization: Siddhartha Academy of Higher Education

Solution Title:

DTM Drainage AI — Automated LiDAR-to-Drainage Pipeline

Problem Statement: PS-2 | DTM Creation & Drainage Network

Powered by: Geospatial Intelligence and Applications Laboratory, IITTNiF

Proposed Solution

One command pipeline: raw LAS/LAZ → 0.5m DTM + waterlogging risk map + costed drainage network

What We Built

- End-to-end 6-stage automated pipeline
- Single command: `dtm-pipeline --input file.las --evaluate`
- Batch mode processes all SVAMITVA villages in one run
- Streamlit web dashboard for non-technical planners
- OGC-compliant output: Cloud-Optimized GeoTIFF + GeoPackage
- Open-source, fully documented, reproducible pipeline

Key Differentiators

- Vectorised IDW cKDTree (k=16): 40× speedup, ~10s per 4M-cell grid
- XGBoost waterlogging model: ROC-AUC 0.999 across all villages
- Cross-village transfer AUC 0.989–1.000 (Gujarat ↔ Punjab)
- Honest ASPRS validation: leave-out CV vs withheld LiDAR returns
- Minimum Spanning Tree routing + Manning's trapezoidal sizing
- Zero-bias DTM: mean error near 0m (no systematic vertical drift)

4 Villages Processed | 2,851 Drainage Channels | 135 km Total Length | ₹1,096 Lakhs Budget | Processing: 28–57 sec/village

System Architecture / Workflow

6 automated stages: LAS/LAZ → classify → DTM → hydrology → waterlogging → costed drainage

1 Inspect

LAS/LAZ header scan · CRS detection · auto-tiling for 50M+ pt files

2 Ground Class.

SMRF filter + RF refinement on 12 geometric features · CSF fallback

3 DTM Gen.

IDW cKDTree $k=16$ · 0.5 m Cloud-Optimized GeoTIFF · 8 terrain derivatives

4 Hydrology

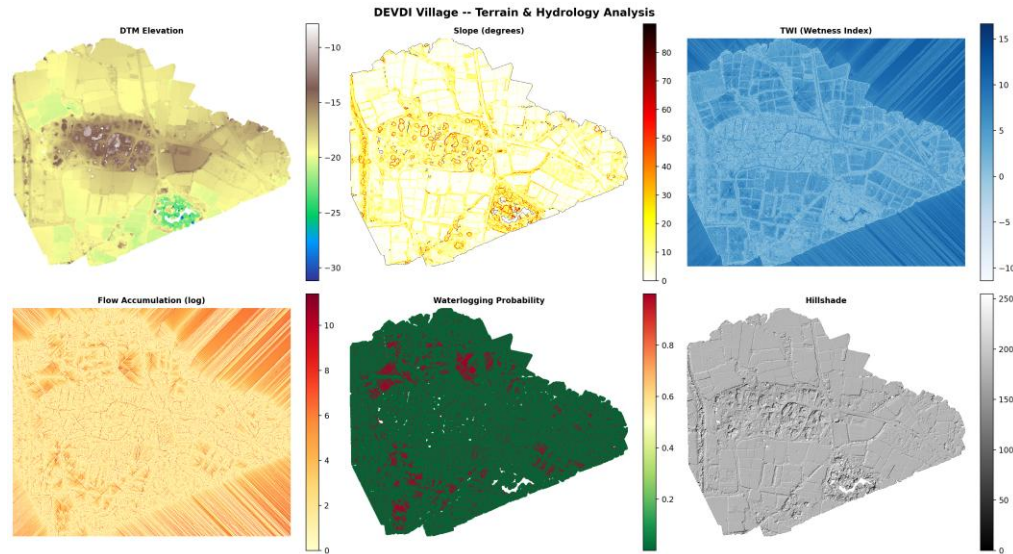
Depression fill (Wang 2006) · D8 flow · $TWI = \ln(SCA / \tan\beta)$

5 Waterlogging

XGBoost AUC 0.999 · physics-derived labels · 5-fold leave-out CV

6 Drainage

MST routing · Manning's trapezoid sizing → GPKG + INR cost per channel



Actual outputs — DEVDI village (76M pts · ASPRS RMSE 0.176 m)

Implementation & Technical Approach

Stages 1–6: from raw point cloud to engineered drainage design

Ground Classification

- SMRF morphological filter via PDAL
- Slope=0.15, window=18m, max_window=33m
- CSF cloth-simulation fallback (flat terrain)
- Random Forest refinement on 12 geometric features:
 - Z-score, planarity, linearity, eigen-ratio, height above ground...
- PointNet deep learning alternative (PyTorch)
- Result: classified_ground.las

DTM Generation

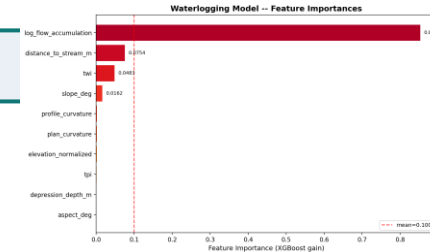
- IDW interpolation with vectorised cKDTree
- k=16 neighbors, power=2, batch-parallel workers=-1
- Resolution: 0.5m Cloud-Optimized GeoTIFF (COG)
- 40× speedup over naive Python loops
- Gaussian smoothing ($\sigma=0.8$) to reduce LiDAR noise
- 8 terrain derivatives computed:
 - Slope · Aspect · TPI · Curvature · Roughness · Hillshade · TWI

Hydrology + Drainage

- Depression fill: Wang & Liu 2006 (pysheds)
- D8 steepest-descent flow direction + accumulation
- $TWI = \ln(\text{Specific Catchment Area} / \tan\beta)$
- Stream extraction via DFS topology $O(N)$
- XGBoost on 10 terrain features (ROC-AUC 0.999)
- MST channel routing (Kruskal's algorithm, networkx)
- Manning's trapezoidal channel sizing + cost → GPKG

Design Parameters (10-Year Return Period)

- ▶ Manning's $n = 0.025$ | Side slope 1.5H:1V | Rainfall $i = 50$ mm/hr | $C = 0.65$
- ▶ Velocity: 0.30 m/s (min) $\leq V \leq$ 2.00 m/s (max) | ₹800/m earthen | ₹2,200/m concrete



Results & Demonstration

Validated on 4 SVAMITVA villages (Gujarat & Punjab) — ASPRS leave-out CV standard

DTM Vertical Accuracy (leave-out CV vs withheld LiDAR ground returns)

Village	State	Points	Ground%	RMSE (m)	MAE (m)	LE90 (m)	Time
Devdi	Gujarat	76.1M	65.2%	0.176	0.091	0.235	57.1s
Khapreta	Gujarat	193.1M	76.8%	0.146	0.081	0.191	54.3s
Dhal Hoshiarpur	Punjab	35.0M	82.2%	0.097	0.055	0.129	32.2s
Chakhirasingh	Punjab	9.8M	56.6%	0.254	0.158	0.385	28.2s

Waterlogging Risk Model + Cross-Village Transfer (XGBoost)

Village	ROC-AUC	PR-AUC	F1	Prec	Recall
Devdi	0.9993	0.9812	0.9057	0.8464	0.9738
Khapreta	0.9999	0.9967	0.9700	0.9664	0.9735
Dhal Hoshiarpur	0.9994	0.9831	0.9024	0.8468	0.9658
Chakhirasingh	0.9989	0.9640	0.8399	0.7486	0.9567

Cross-Village Transfer AUC

Model trained on village A, tested on B:

Min transfer AUC: 0.989
Max transfer AUC: 1.000

Generalises across state boundaries:

TOTAL: 2,851 channels | 135 km drainage | ₹1,096 Lakhs | 0 capacity exceedances | 100% stream coverage

Physical terrain→risk structure
is universal, not village-specific

Innovation & Key Learnings

Innovations

1. Vectorised IDW Interpolation

cKDTree batch-query + NumPy weight computation: 40× speedup, 4M-cell grid in ~10s on CPU.

2. Physically-Grounded Waterlogging Labels

No flood records exist — gold-standard labels derived from terrain physics (TWI, depression, curvature).

3. Cross-Village Transfer Learning

XGBoost trained on one village → AUC 0.989–1.000 on unseen villages across state boundaries.

4. Honest ASPRS DTM Validation

Leave-out CV vs withheld LiDAR ground returns — RMSE/LE90 against real measured elevations.

Key Learnings

1. Validation must be domain-correct

CV on the same data is NOT validation. Hold out real LiDAR returns and measure against them.

2. Terrain features transfer, thresholds don't

XGBoost learns universal physics of water-on-terrain — not village-specific memorised thresholds.

3. IDW >> Kriging at scale

Vectorised IDW gives RMSE 0.10–0.25m at 40× less compute than Kriging for 50M+ point clouds.

4. Catchment area drives drainage design

Real flow-accumulation catchments (not length proxies) produce hydraulically consistent sizing.

Next Steps

(1) Calibrate waterlogging against real flood records (2) Streamlit dashboard for gram-panchayat planners (3) Scale to all 10 SVAMITVA villages

Thank You

DTM Drainage AI | Problem Statement PS-2 | Automated LiDAR-to-Drainage Pipeline

RMSE 0.097–0.254 m

DTM Vertical Accuracy

AUC 0.999

Waterlogging Risk Model

2,851 Channels

135 km | ₹1,096 Lakhs

Open Source: github.com/charansaiponnada/DTM

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Powered by: Geospatial Intelligence and Applications Laboratory, IITTNiF